



An Optimal Double Stopping Rule for a Buying-Selling Problem

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Abstract

We consider a buying-selling problem with a finite time horizon when two stops of a sequence of dependent observations can be made. The aim is to find an optimal sequential procedure which maximizes the total expected revenue. In this paper, we obtain an optimal double stopping rule and apply it for a geometric random walk and an autoregressive sequence.

Keywords Sequential decision analysis · Optimal stopping rules · Buying-selling problem · Geometric random walk · Autoregressive sequence

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1 Introduction

Let $y_1, y_2, \dots, y_N$ be a sequence of random variables. The random variable y_n can be interpreted as a value of asset at time n . We observe these random variables sequentially and have to decide when we must stop. The first stopping means a buying an asset, the second one signifies a selling an asset. Our decision to stop depends on the observations already made, but does not depend on the future which is not yet known. After two stoppings at times m_1, m_2 , $1 \leq m_1 < m_2 \leq N$ we get a gain $Z_{m_1, m_2} = y_{m_2} - y_{m_1}$. If we do not buy anything until time N then we will obtain the gain $Z_{m_1, m_2} = 0$, we may assume that $m_1 = N$, $m_2 = N + 1$. This implies that

$$1 \leq m_1 \leq N_1, \quad N_1 = N, \\ m_1 < m_2 \leq N_2(m_1), \quad N_2(m_1) = \begin{cases} N & \text{if } m_1 < N, \\ N + 1 & \text{if } m_1 = N. \end{cases}$$

The problem consists of finding a procedure for maximizing the expected gain.

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Different types of selling problems, which are also called house-selling problems (see Albright 1977), have been considered by many authors. The problem is related to the well-known Cayley's problem (see, for example, Ferguson 1989; Moser 1956). In MacQueen and Miller (1960), the authors consider the problem of selling several identical objects when single offers come in daily and recall of past offers. Selling non-identical objects with one offer per time period is discussed in Albright (1974). A generalisation of the house-selling problem with vector offers is studied in Bruss and Ferguson (1997). In Rosenfield et al. (1983), the authors consider both the case where only the most recently obtained offer may be accepted (no recall), and the case where any previously received offer may be accepted (recall allowed) under both finite and infinite horizons. Asset-selling problems with holding costs are proposed in David and Levi (2001). A non-parametric approach to a buying-selling problem with a finite time horizon is proposed in Chun et al. (2002).

Various problems have been considered within the framework of multiple stopping rules. A multi-person stopping game with players' priorities and multiple stopping is considered in Krasnosielska-Kobos (2016). In Targino et al. (2017), the authors study a class of insurance products where the policy holder has the right to insure several of its annual operational losses in a finite time horizon. In Nikolaev and Sofronov (2007), a problem of selling several identical objects with a finite horizon and with one offer per time period and no recall of past offers is presented. An effective number of stops for a generalisation of the problem with a fixed number of offers per time period is derived in Sofronov (2013). In Sofronov (2016), we consider a buying-selling problem with the finite time horizon when several stops can be made. In this paper, we generalise a buying-selling problem (Sofronov et al. 2006) to a case of dependent random variables.

Many other applications of sequential decision analysis can be found in the literature: job search (Lippman and McCall 1976; Stein et al. 2003), engineering (David 1998; Righter 1987), resource allocation (Papastavrou et al. 1996), behavioural ecology (Dombrovsky and Perrin 1994; Hutchinson and Halupka 2004; Real 1990, 1991), and insurance (Karpowicz and Szajowski 2007; Targino et al. 2017),

The paper is structured as follows. Section 2 includes necessary results of the theory of optimal double stopping rules. In Section 3, we find a double optimal stopping rule for the buying-selling problem. Sections 4 and 5 present optimal stopping rules for two important types of sequences: a geometric random walk and an autoregressive process of order 1.

2 Optimal Double Stopping Rules

Assume we sequentially observe random variables $y_1, y_2, \dots$, and we are allowed to stop two times and choose the corresponding observations y_{m_1} and y_{m_2} . We stop at times m_1 and m_2 after having observed $y_1, \dots, y_{m_1}$, and $y_1, \dots, y_{m_2}$, respectively. After two stops at times m_1 and m_2 , we receive a gain Z_{m_1, m_2} , which is a function of $y_1, \dots, y_{m_2}$. In the buying-selling problem we consider in this paper, the gain function takes the following form:

$$Z_{m_1, m_2} = \begin{cases} y_{m_2} - y_{m_1}, & 1 \leq m_1 < m_2 \leq N, \\ 0, & m_1 = N, m_2 = N + 1. \end{cases} \quad (1)$$

Other examples of the gain function can be found in Nikolaev and Sofronov (2007), Sofronov (2013), Sofronov (2016), and Targino et al. (2017).

Formally, the mathematical framework consists of

- (a) a probability space $(\Omega, \mathcal{F}, \mathbf{P})$;

- (b) a non-decreasing sequence $\{\mathcal{F}_n, n \geq 1\}$ of the σ -algebras such that $\mathcal{F}_{m_1} \subseteq \mathcal{F}_{m_2} \subseteq \mathcal{F}$ for all $1 \leq m_1 < m_2$;
- (c) a random process $\{Z_{m_1, m_2}, \mathcal{F}_{m_2}, m_2 > m_1\}$, Z_{m_1, m_2} is $\mathcal{F}_{m_2}$ -measurable, for each fixed integer $m_1 \geq 1$.

Within this framework, the decision to stop at times m_1 and m_2 depends solely on the values of the random sequence already observed and not on any future values. We also assume that it is not possible to stop twice at the same discrete time, that is, $1 \leq m_1 < m_2$. The following definition of a multiple stopping rule reflects these assumptions; for further details on the theory of multiple optimal stopping rules, see Nikolaev (1979), Nikolaev (1981), Nikolaev (1999), Nikolaev and Sofronov (2007), and Sofronov et al. (2006).

Definition 1 A pair of integer-valued random variables $\tau = (\tau_1, \tau_2)$ is called a *double stopping rule* if the following conditions hold:

- (a) $\{\omega \in \Omega : \tau_1(\omega) = m_1\} \in \mathcal{F}_{m_1}$, $\{\omega \in \Omega : \tau_1(\omega) = m_1, \tau_2(\omega) = m_2\} \in \mathcal{F}_{m_2}$ for all $m_2 > m_1 \geq 1$;
- (b) $1 \leq \tau_1 < \tau_2 < +\infty$ ($\mathbf{P}$ -a.s.).

We will denote the conditional expectations relative to the σ -algebras $\mathcal{F}_n$ by $\mathbf{E}_n$. The following condition needed for the existence of all considered expectations (see Haggstrom 1967; Nikolaev 1979, 1981; Stadje 1985) will be assumed throughout this section. Other conditions that ensure the existence of the expectations are discussed in Kösters (2004).

$$\mathbf{E}\left(\sup_{m_1} \mathbf{E}_{m_1}(\sup Z_{m_1, m_2}^+)\right) < +\infty,$$

where $x^+ = \max\{x, 0\}$.

Using the mathematical definition of a double stopping rule, we can now define optimal double stopping rules.

Definition 2 Let S_m be the class of double stopping rules $\tau = (\tau_1, \tau_2)$ such that $\tau_1 \geq m$ ($\mathbf{P}$ -a.s.). The function

$$v_m = \sup_{\tau \in S_m} \mathbf{E}Z_\tau$$

is called the *m-value*. In particular, if $m = 1$ then $v = v_1$ is the *expected gain* (or the *value of the game*).

Thus, S_m is the set of all double stopping times with respect to $\{\mathcal{F}_n\}$ such that $(\tau_1, \tau_2) \in S_m$ if $\tau_1 \geq m$.

Definition 3 A double stopping rule $\tau^* \in S_m$ is called an *optimal double stopping rule* in S_m if $\mathbf{E}Z_{\tau^*}$ exists and $\mathbf{E}Z_{\tau^*} = v_m$.

Assume we observe a finite sequence of random variables $y_1, y_2, \dots, y_N$. Let

$$\{Z_{m_1, m_2}, 1 \leq m_1 \leq N_1, m_1 < m_2 \leq N_2(m_1)\}$$

be a family of random variables, where N_1 and $N_2(m_1)$ are natural numbers. The sequences $\{X_{m_1}\}$, $\{V_{m_1}\}$ and $\{V_{m_1, m_2}\}$ are needed for constructing the stopping rules. They can be

defined by backward induction from the following recurrent equations (see Chow et al. 1971; Haggstrom 1967):

$$V_{m_1, m_2} = \max\{Z_{m_1, m_2}, \mathbf{E}_{m_2} V_{m_1, m_2+1}\}, \quad (2)$$

$$V_{m_1, N_2(m_1)} = Z_{m_1, N_2(m_1)}, \quad (3)$$

$$X_{m_1} = \mathbf{E}_{m_1} V_{m_1, m_1+1}, \quad (4)$$

$$V_{m_1} = \max\{X_{m_1}, \mathbf{E}_{m_1} V_{m_1+1}\}, \quad (4)$$

$$V_{N_1} = X_{N_1},$$

for $1 \leq m_1 < N_1, m_1 < m_2 \leq N_2(m_1)$. Here V_{m_1, m_2} represents the maximum gain that is possible to receive at time m_2 . If $m_2 = N_2(m_1)$, then the gain is $V_{m_1, N_2(m_1)} = Z_{m_1, N_2(m_1)}$. If $m_2 < N_2(m_1)$, we can either stop or continue. If we stop, the gain is Z_{m_1, m_2} , and if we continue, the gain is $\mathbf{E}_{m_2} V_{m_1, m_2+1}$. After having observed the first m_1 observations, X_{m_1} can be interpreted as the conditional expected gain if we stop for the first time at m_1 and use an optimal stopping time for the second stage. The interpretation of V_{m_1} is similar to V_{m_1, m_2} , being the maximum gain that is possible to receive at time m_1 .

Theorem 1 (Sofronov 2016) and Theorem 2 (Sofronov 2016) give the existence conditions and the structure of an optimal multiple stopping rule in S_m and the characterization of the m -value v_m by means of the sequence $\{V_m\}$, respectively. The proofs of the theorems can be found in Nikolaev (1979). Using these results, we obtain the optimal double stopping rule τ^* and the value v_m .

Assume we observe a sequence of independent random variables $y_1, \dots, y_N$ and that the gain function is as in Eq. 1. The following result presented in Sofronov et al. (2006) gives the optimal double stopping rule that maximizes the expected gain $\mathbf{E}Z_{m_1, m_2}$.

Theorem 1 *Let $y_1, y_2, \dots, y_N$ be a sequence of independent random variables with known distribution functions, and the gain function Z_{m_1, m_2} be defined in Eq. 1. Let $v^{L, l}$ be the value of a game with l ($l \leq 2$) stops and L ($L \leq N$) steps. If there exist $\mathbf{E}y_1, \mathbf{E}y_2, \dots, \mathbf{E}y_N$, then the value $v = v^{N, 2}$, where*

$$v^{n, 2} = \mathbf{E}(\max\{v^{n-1, 1} - y_{N-n+1}, v^{n-1, 2}\}), \quad 2 \leq n \leq N, \quad v^{1, 2} = 0,$$

$$v^{n, 1} = \mathbf{E}(\max\{y_{N-n+1}, v^{n-1, 1}\}), \quad 1 \leq n \leq N-1, \quad v^{0, 1} = -\infty.$$

We put

$$\tau_1^* = \min\{\min\{m_1 : 1 \leq m_1 \leq N-1, y_{m_1} \leq v^{N-m_1, 1} - v^{N-m_1, 2}\}, N\},$$

$$\tau_2^* = \min\{\min\{m_2 : \tau_1^* < m_2 \leq N, y_{m_2} \geq v^{N-m_2, 1}\}, \{N+1 : \tau_1^* = N\}\},$$

then $\tau^* = (\tau_1^*, \tau_2^*)$ is the optimal double stopping rule.

3 An Optimal Stopping Rule for the Buying-selling Problem

From Theorem 1, we can easily see that if we observe a sequence of independent random variables, then in order to find the optimal rule we actually only need to calculate unconditional expectations of the form $\mathbf{E}(\max\{c_1 + c_2 y, c_3\})$, where c_1, c_2, c_3 are constants. This means that the values $v^{n, 1}$ and $v^{n, 2}$ can be calculated in advance, before we start observing the sequence of random variables. If the assumption of independence is not used, from Eqs. 2, 3 and 4, we can see that we need to calculate conditional expectations so that the

$v^{n,1}$ and $v^{n,2}$ are functions of the previous observations. The following theorem generalises Theorem 1 to a case of dependent random variables.

Theorem 2 Let $\{y_n\}$, Z_{m_1, m_2} be as introduced in Section 1. Let $v^{L,l}$ be the value of a game with l ($l \leq 2$) stoppings and L ($L \leq N$) steps; then the value $v = v^{N,2}$, where

$$\begin{aligned} v^{n,2} &= E_{N-n}(\max\{v^{n-1,1} - y_{N-n+1}, v^{n-1,2}\}), \quad 2 \leq n \leq N, \quad v^{1,2} = 0, \\ v^{n,1} &= E_{N-n}(\max\{y_{N-n+1}, v^{n-1,1}\}), \quad 1 \leq n \leq N-1, \quad v^{0,1} = -\infty. \end{aligned}$$

We define

$$\begin{aligned} \tau_1^* &= \min\{\min\{m_1 : 1 \leq m_1 \leq N-1, y_{m_1} \leq v^{N-m_1,1} - v^{N-m_1,2}\}, N\}, \\ \tau_2^* &= \min\{\min\{m_2 : \tau_1^* < m_2 \leq N, y_{m_2} \geq v^{N-m_2,1}\}, \{N+1 : \tau_1^* = N\}\}, \end{aligned}$$

then $\tau^* = (\tau_1^*, \tau_2^*)$ is the optimal double stopping rule.

Proof If $m_1 = N$, $m_2 = N+1$, then $Z_{m_1, m_2} = 0$. It follows easily that $v^{1,2} = 0$.

Let us consider the case $1 \leq m_1 \leq N-1$, $m_1 < m_2 \leq N$. From Eq. 2 we obtain

$$\begin{aligned} V_{m_1, m_2} &= \max\{y_{m_2} - y_{m_1}, E_{m_2} V_{m_1, m_2+1}\} \\ &= \max\{y_{m_2}, E_{m_2} V_{m_1, m_2+1} + y_{m_1}\} - y_{m_1} \\ &= \max\{y_{m_2}, v^{N-m_2,1}\} - y_{m_1} \end{aligned} \quad (5)$$

with

$$v^{N-m_2+1,1} = E_{m_2-1}(\max\{y_{m_2}, v^{N-m_2,1}\}), \quad m_1 < m_2 < N,$$

where $v^{N-m_2+1,1}$ may depend on $y_1, \dots, y_{m_2-1}$. If $m_2 = N$, then $v^{N-m_2+1,1} = v^{1,1} = E_{N-1}y_N$ since $v^{0,1} = -\infty$.

From Eqs. 3 and 5 we have

$$X_{m_1} = E_{m_1}(\max\{y_{m_1+1}, v^{N-m_1-1,1}\} - y_{m_1}) = v^{N-m_1,1} - y_{m_1}. \quad (6)$$

From Eqs. 4 and 6 we have

$$\begin{aligned} V_{m_1} &= \max\{v^{N-m_1,1} - y_{m_1}, E_{m_1} V_{m_1+1}\} \\ &= \max\{v^{N-m_1,1} - y_{m_1}, v^{N-m_1,2}\} \end{aligned} \quad (7)$$

with

$$v^{N-m_1+1,2} = E_{m_1-1}(\max\{v^{N-m_1,1} - y_{m_1}, v^{N-m_1,2}\}), \quad m_1 < N-1,$$

where $v^{N-m_1+1,2}$ may depend on $y_1, \dots, y_{m_1-1}$. If $m_1 = N-1$, then

$$v^{N-m_1+1,2} = v^{2,2} = E_{N-2}(\max\{v^{1,1} - y_{N-1}, 0\})$$

since $v^{1,2} = 0$.

Taking into account Theorem 2 (Sofronov 2016), we obtain the value $v = v^{N,2}$.

Now, using Theorem 1 (Sofronov 2016) together with Eqs. 5, 6 and 7, we obtain the optimal double stopping rule $\tau^* = (\tau_1^*, \tau_2^*)$:

$$\begin{aligned} \tau_1^* &= \min\{\min\{m_1 : 1 \leq m_1 \leq N-1, y_{m_1} \leq v^{N-m_1,1} - v^{N-m_1,2}\}, N\}, \\ \tau_2^* &= \min\{\min\{m_2 : \tau_1^* < m_2 \leq N, y_{m_2} \geq v^{N-m_2,1}\}, \{N+1 : \tau_1^* = N\}\}. \end{aligned}$$

This completes the proof. $\square$

4 An Optimal Stopping Rule for a Geometric Random Walk

Consider a sequence $y_1, y_2, \dots, y_N$,

$$y_n = y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_n}, \quad y_0 \in \{\lambda^i, i = 0, \pm 1, \dots\}, \quad \lambda > 1,$$

where ε_i are independent and identically distributed random variables with $P(\varepsilon_i = 1) = p$ and $P(\varepsilon_i = -1) = q$, $p + q = 1$, $0 < p < 1$. This sequence, which plays an important role in stochastic financial modelling, is called a geometric random walk (Shiryaev 1999, Chapter II, §1e).

Using Theorem 2, we will find an optimal double stopping rule for the sequence $y_1, y_2, \dots, y_N$. We have, for $m_1 < m_2 < N$,

$$v^{N-m_2+1,1} = \mathbf{E}_{m_2-1}(\max\{y_{m_2}, v^{N-m_2,1}\})$$

where $\mathbf{E}_m$ is the conditional expectation with respect to $\mathcal{F}_m = \sigma(y_1, \dots, y_m)$, σ -algebra generated by random variables $y_1, \dots, y_m$. For $m_2 = N$, taking into account that $v^{0,1} = -\infty$, we have

$$v^{1,1} = \mathbf{E}_{N-1} y_N = r y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-1}}, \quad r = \lambda p + \lambda^{-1} q.$$

Then

$$\begin{aligned} v^{2,1} &= \mathbf{E}_{N-2}(\max\{y_{N-1}, v^{1,1}\}) \\ &= \mathbf{E}_{N-2}(\max\{y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-1}}, r y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-1}}\}) \\ &= \mathbf{E}_{N-2}(\max\{\lambda^{\varepsilon_{N-1}}, r \lambda^{\varepsilon_{N-1}}\}) y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-2}} \\ &= \begin{cases} r y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-2}}, & r \leq 1 \\ r^2 y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-2}}, & r > 1. \end{cases} \end{aligned}$$

This means that the decision-maker should stop if $r \leq 1$, and the decision-maker should continue if $r > 1$. In the same way,

$$\begin{aligned} v^{3,1} &= \mathbf{E}_{N-3}(\max\{y_{N-2}, v^{2,1}\}) \\ &= \begin{cases} \mathbf{E}_{N-3}(\max\{\lambda^{\varepsilon_{N-2}}, r \lambda^{\varepsilon_{N-2}}\}) y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-3}}, & r \leq 1 \\ \mathbf{E}_{N-3}(\max\{\lambda^{\varepsilon_{N-2}}, r^2 \lambda^{\varepsilon_{N-2}}\}) y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-3}}, & r > 1 \end{cases} \\ &= \begin{cases} r y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-3}}, & r \leq 1 \\ r^3 y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-3}}, & r > 1. \end{cases} \end{aligned}$$

Continuing this process, we can show that

$$v^{n,1} = \begin{cases} r y_{N-n}, & r \leq 1 \\ r^n y_{N-n}, & r > 1, \end{cases} \quad 1 \leq n \leq N-1.$$

Similarly,

$$\begin{aligned} v^{2,2} &= \mathbf{E}_{N-2}(\max\{v^{1,1} - y_{N-1}, 0\}) \\ &= \mathbf{E}_{N-2}(\max\{(r-1) y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-1}}, 0\}) \\ &= \begin{cases} 0, & r \leq 1 \\ (r^2 - r) y_0 \lambda^{\varepsilon_1 + \dots + \varepsilon_{N-2}}, & r > 1. \end{cases} \end{aligned}$$

Then

$$\begin{aligned} v^{3,2} &= \mathbf{E}_{N-3}(\max\{v^{2,1} - y_{N-2}, v^{2,2}\}) \\ &= \begin{cases} \mathbf{E}_{N-3}(\max\{(r-1)\lambda^{\varepsilon_{N-2}}, 0\})y_0\lambda^{\varepsilon_1+\dots+\varepsilon_{N-3}}, & r \leq 1 \\ \mathbf{E}_{N-3}(\max\{(r^2-1)\lambda^{\varepsilon_{N-2}}, (r^2-r)\lambda^{\varepsilon_{N-2}}\})y_0\lambda^{\varepsilon_1+\dots+\varepsilon_{N-3}}, & r > 1 \end{cases} \\ &= \begin{cases} 0, & r \leq 1 \\ (r^3-r)y_0\lambda^{\varepsilon_1+\dots+\varepsilon_{N-3}}, & r > 1. \end{cases} \end{aligned}$$

It can be proved by induction that

$$v^{n,2} = \begin{cases} 0, & r \leq 1 \\ (r^n-r)y_{N-n}, & r > 1, \end{cases} \quad 2 \leq n \leq N.$$

with the value $v = v^{N,2}$.

This implies that the optimal decision rule depends solely on r , which is a known value. Therefore, the rule will be trivial: if $r \leq 1$, do not buy anything; otherwise, if $r > 1$, buy in the beginning and sell in the end of the time interval.

5 An Optimal Stopping Rule for an AR(1) Process

Consider an AR(1) sequence $y_1, y_2, \dots, y_N$,

$$y_n = \alpha y_{n-1} + \varepsilon_n, \quad y_0 = 0,$$

where $\varepsilon_1, \dots, \varepsilon_N$ are independent random variables having normal distribution $N(\mu, \sigma^2)$. Assume that $\alpha \in (-1, 1)$, then the AR(1) model is stationary.

Following the proof of Theorem 2, we have, for $m_1 < m_2 < N$,

$$v^{N-m_2+1,1} = \mathbf{E}_{m_2-1}(\max\{y_{m_2}, v^{N-m_2,1}\})$$

where $\mathbf{E}_m$ is the conditional expectation with respect to $\mathcal{F}_m = \sigma(y_1, \dots, y_m)$, σ -algebra generated by random variables $y_1, \dots, y_m$. If $m_2 = N$, $v^{N-m_2+1,1} = v^{1,1} = \mathbf{E}_{N-1}y_N = \mu + \alpha y_{N-1}$ since $v^{0,1} = -\infty$. Note that $v^{N-m+1,i}$, $i = 1, 2$, is a function of y_{m-1} , $2 \leq m \leq N$.

Similarly, for $m_1 < N-1$,

$$v^{N-m_1+1,2} = \mathbf{E}_{m_1-1}(\max\{v^{N-m_1,1} - y_{m_1}, v^{N-m_1,2}\}).$$

If $m_1 = N-1$, taking into account that $v^{1,2} = 0$, we have

$$\begin{aligned} v^{2,2} &= \mathbf{E}_{N-2}(\max\{v^{1,1} - y_{N-1}, 0\}) \\ &= \mathbf{E}_{N-2}(\max\{\mu + \alpha y_{N-1} - y_{N-1}, 0\}), \end{aligned}$$

where the conditional distribution of y_{N-1} given y_{N-2} is Normal with mean $\mu + y_{N-2}$ and variance σ^2 .

We see that in this case the $v^{n,1}$ and $v^{n,2}$ are functions of the previous observations. Note that for a buying-selling problem with independent observations the corresponding values do not depend on the past observations and, therefore, can be calculated in advance; see Sofronov et al. (2006).

In order to find the functions $v^{n,1} = v^{n,1}(t)$ and $v^{n,2} = v^{n,2}(t)$, we propose to use numerical integration methods. To do this, for each n , we evaluate the functions at a finite set of points, $t_1, t_2, \dots, t_K$. Since in our case the integration is on the whole real line, we

will use the Gauss-Hermite quadrature (see, for example, (Abramowitz and Stegun 1972, p. 890, eq. 25.4.46)):

$$\int_{-\infty}^{\infty} g(x) e^{-x^2} dx = \sum_{i=1}^K w_i g(x_i) + R_K,$$

where x_i is the i -th zero of the Hermite polynomial $H_K(x)$, $i = 1, 2, \dots, K$, the weights w_i are given by

$$w_i = \frac{2^{K-1} K! \sqrt{\pi}}{K^2 (H_{K-1}(x_i))^2},$$

and the remainder is

$$R_K = \frac{K! \sqrt{\pi}}{2^K (2K)!} g^{(2K)}(\xi), \quad -\infty < \xi < \infty.$$

Let

$$f_{t_k}(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu-\alpha t_k)^2/(2\sigma^2)}$$

be the probability density function of the normal distribution $N(\mu + \alpha t_k, \sigma^2)$. Then it can easily be shown that

$$\int_{-\infty}^{\infty} g(x) f_{t_k}(x) dx = \frac{1}{\sqrt{\pi}} \sum_{i=1}^K w_i g(\sqrt{2}\sigma x_i + \mu + \alpha t_k) + R_K.$$

Denote

$$\begin{aligned} g_{t_k}^{n,1}(x) &= \max\{x, v^{n,1}(x)\}, \\ g_{t_k}^{n,2}(x) &= \max\{v^{n,1}(x) - x, v^{n,2}(x)\}. \end{aligned}$$

Then, for $k = 1, 2, \dots, K$,

$$\begin{aligned} v^{1,1}(t_k) &= \mu + \alpha t_k, \\ v^{n,1}(t_k) &= \int_{-\infty}^{\infty} g_{t_k}^{n-1,1}(x) f_{t_k}(x) dx \\ &= \frac{1}{\sqrt{\pi}} \sum_{i=1}^K w_i g_{t_k}^{n-1,1}(\sqrt{2}\sigma x_i + \mu + \alpha t_k), \\ &n = 2, \dots, N-1. \end{aligned} \tag{8}$$

Similarly, for $k = 1, 2, \dots, K$,

$$\begin{aligned} v^{1,2}(t_k) &= 0, \\ v^{n,2}(t_k) &= \int_{-\infty}^{\infty} g_{t_k}^{n-1,2}(x) f_{t_k}(x) dx \\ &= \frac{1}{\sqrt{\pi}} \sum_{i=1}^K w_i g_{t_k}^{n-1,2}(\sqrt{2}\sigma x_i + \mu + \alpha t_k), \\ &n = 2, \dots, N. \end{aligned} \tag{9}$$

Though it is possible to choose the set of points $t_1, t_2, \dots, t_K$ arbitrarily, we can choose the points in such a way that the set is identical for all values of n . In this case, $t_k = \sqrt{2}\sigma x_k + \mu + \alpha t_k$, $k = 1, 2, \dots, K$. Then, for $\alpha \neq 1$,

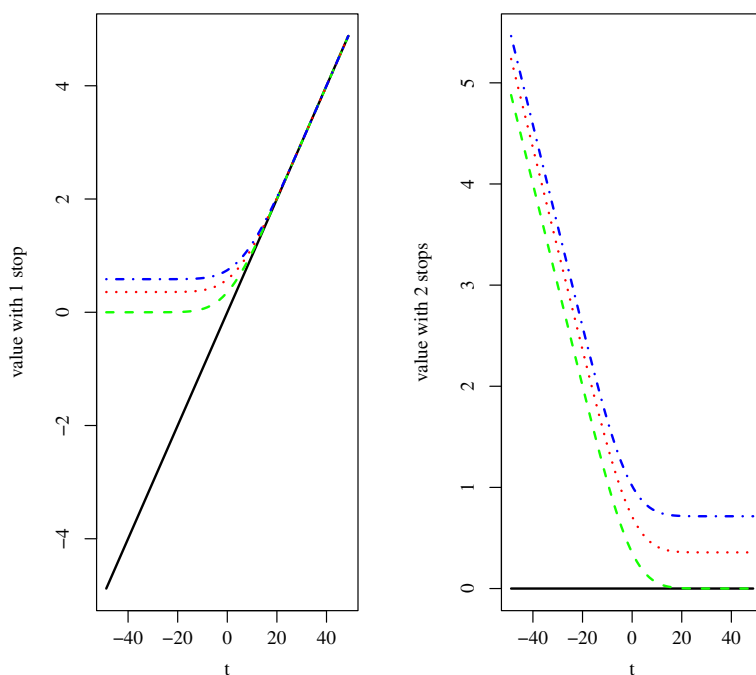
$$t_k = \frac{\sqrt{2}\sigma x_k + \mu}{1 - \alpha}, \quad k = 1, 2, \dots, K.$$

Table 1 The values $v^{n,1}$ and $v^{n,2}$ compared to the values obtained by the Gauss-Hermite quadrature, $v_g^{n,1}$ and $v_g^{n,2}$, with $K = 500$, $N = 7$, $\text{Normal}(0, 1)$

n	$v^{n,1}$	$v_g^{n,1}$	Error	$v^{n,2}$	$v_g^{n,2}$	Error
1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.3989	0.3992	0.0003	0.3989	0.3992	−0.0003
3	0.6297	0.6301	−0.0004	0.7979	0.7985	−0.0006
4	0.7904	0.7902	0.0002	1.1184	1.1191	−0.0007
5	0.9126	0.9123	0.0003	1.3746	1.3749	−0.0003
6	1.0108	1.0101	0.0007	1.5844	1.5845	−0.0001
7	1.0924	1.0919	0.0005	1.7604	1.7605	−0.0001

This allows us to sequentially evaluate the sets of values $v^{n,1}(t_k)$ and $v^{n,2}(t_k)$, $k = 1, 2, \dots, K$, using the previous layer of values $v^{n-1,1}(t_k)$ and $v^{n-1,2}(t_k)$; see Eqs. 8 and 9. If $\alpha = 1$, similar to the case of the geometric random walk considered in Section 4, the optimal rule is trivial (see Sofronov 2016, Section 4).

In the examples below, in order to compute the quadrature nodes x_i and the weights w_i , we use the function `gaussHermiteData` (Blocker 2014). The function `approxfun` (R Core Team 2018) is used to perform linear interpolation of the pairs $(t_k, v^{n,1}(t_k))$ and $(t_k, v^{n,2}(t_k))$, $k = 1, 2, \dots, K$.

**Fig. 1** The values $v^{n,1}$ (left) and $v^{n,2}$ (right) for $n = 1$ (black solid line), $n = 2$ (green dashed line), $n = 3$ (red dotted line), and $n = 4$ (blue dot-dashed line)

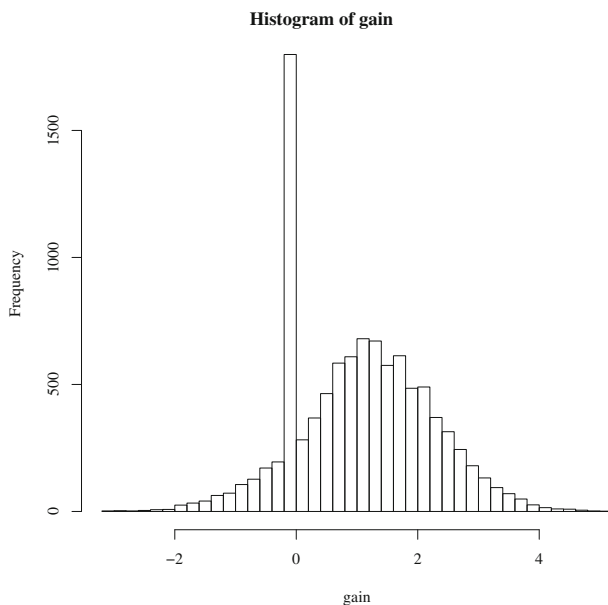


Fig. 2 The histogram of the gains when the double optimal stopping rule was applied to 10,000 simulated sequences

5.1 Example 1

Table 1 shows the values $v^{n,1}$ and $v^{n,2}$, $n = 1, 2, \dots, 7$, for a sequence with parameters $\alpha = 0$ (no dependence), $\mu = 0$, and $\sigma^2 = 1$. Since we observe a sequence of independent random variables, we can use Theorem 1 to calculate the $v^{n,1}$ and $v^{n,2}$ (for further details, see Sofronov et al. 2006, Example 3) and compare these values with the approximate values $v_g^{n,1}$ and $v_g^{n,2}$ obtained by the Gauss-Hermite quadrature (8) and (9). The small values of the errors in the table illustrate a good accuracy of the numerical procedure.

5.2 Example 2

Figure 1 shows the values $v^{n,1}$ and $v^{n,2}$, $n = 1, 2, 3, 4$, for a sequence with parameters $\alpha = 0.1$, $\mu = 0$, and $\sigma^2 = 1$. The Gauss-Hermite quadrature was applied with parameter $K = 500$.

Using the values of the parameters α , μ , and σ^2 specified above, we generated 10,000 sequences of length $N = 4$ with $y_0 = 0$ and applied the optimal double stopping rule. The

Table 2 The average gains for different rules: optimal (Rule 1), deterministic (Rules 2 and 3) and random (Rule 4)

	Rule 1	Rule 2	Rule 3	Rule 4
Average gain	0.0007	−0.0020	0.0000	−0.0018

estimated average gain of 1.030 is similar to the expected gain $v^{4,2} = 1.015$. Figure 2 displays the histogram of the values of the gain with the spike at 0 indicating a high proportion of cases when we did not buy anything.

For example, if we observe the sequence

$$\{y_n : n = 1, 2, 3, 4\} : -0.1206, -0.1193, 2.2498, -0.1196,$$

then $m_1 = 2$ (because $y_2 = -0.1193 \leq v^{2,1}(y_2) - v^{2,2}(y_2) = 0.3489 - 0.3609 = -0.0120$) and $m_2 = 3$ (because $y_3 = 2.2498 \geq v^{1,1}(y_3) = 0.2250$). This yields the gain $2.2498 - (-0.1193) = 2.3691$.

5.3 Example 3

In Nagaraja et al. (2011), it was shown that AR(1) processes can successfully be applied to house log prices modelling. Assume $\alpha = 0.99$, $\mu = 0$, and $\sigma^2 = 0.002$. Using these values, we generated 10,000 sequences of length $N = 5$ with $y_0 = 0$ and applied the rules defined as follows:

Rule 1 (optimal): the optimal double stopping rule with the Gauss-Hermite quadrature ($K = 500$);

Rule 2 (deterministic): always buy at the beginning ($m_1 = 1$) and sell at the end ($m_2 = 5$) of the time interval;

Rule 3 (deterministic): do not buy anything.

Rule 4 (random): stop randomly at m_1 and m_2 , subject to $1 \leq m_1 < m_2 \leq 5$.

Table 2 displays the average gains of the rules indicating that Rule 1 (the optimal rule) outperforms the other three rules.

6 Conclusion

In this paper, we have studied a buying-selling sequential problem with a finite horizon. We have derived an optimal double stopping rule for a general case of dependent observations. We have also found the optimal rules for types of sequences popular in financial modelling: a geometric random walk and an AR(1) process. To find the optimal rules, we have proposed to use numerical integration methods, which are numerically stable and fast.

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